

UK ROAD SAFETY ANALYSIS (2024) USING POWER BI

Submitted By

Name: Abhisha JL

Course: Data Analytics with GenAI

Guide: Mrs. Rohini

Batch: 11:30 a.m - 1:30 p.m

Table of Contents

1. Introduction
2. Problem Statement
3. Objectives
4. Dataset Description
5. Methodology
6. Data Analysis and Interpretation
7. Findings
8. Conclusion

1. Introduction

Road safety is one of the most significant public safety concerns, as road traffic accidents lead to fatalities, injuries, and economic losses every year. Analyzing accident data helps identify the factors contributing to road collisions and supports the development of effective road safety measures. Business Intelligence (BI) tools such as Microsoft Power BI enable users to transform large datasets into interactive dashboards, making it easier to identify patterns and trends.

This project, "UK Road Safety Analysis (2024) using Power BI," focuses on analyzing road accident data from the United Kingdom (UK) for the year 2024. The dataset, provided in Microsoft Excel (.xlsx) format, was supplied by the faculty for academic analysis. The data was imported into Power BI Desktop, where it was cleaned, transformed, and analyzed using Power Query and various visualization techniques.

The dashboard presents key performance indicators (KPIs), including total collisions, total vehicles involved, total casualties, fatal casualties, serious casualties, and slight casualties. It also provides detailed analysis based on monthly collision trends, weather conditions, light conditions, road surface conditions, vehicle types, and casualty distribution across different age groups.

The interactive dashboard allows users to filter and explore the data efficiently, enabling a better understanding of accident patterns in the UK during 2024. The insights obtained from this analysis can support awareness of road safety trends and demonstrate how Power BI can be used to convert raw data into meaningful visual information for data-driven decision-making.

2. Problem Statement

Road traffic accidents are a major public safety concern, resulting in injuries, fatalities, and significant economic losses. Understanding the causes and patterns of road accidents is essential for improving road safety and supporting informed decision-making. However, accident data is often available in large datasets, making it difficult to identify meaningful trends and insights through manual analysis.

The UK Road Safety 2024 dataset contains information on road collisions, casualties, vehicle types, weather conditions, road surface conditions, light conditions, and age groups. Without proper analysis and visualization, it is challenging to identify the factors associated with road accidents and understand how these factors influence casualty rates and collision patterns.

The problem addressed in this project is the need to transform raw road accident data into an interactive and easy-to-understand dashboard using Microsoft Power BI. By cleaning, organizing, and visualizing the data, the project aims to highlight key road safety indicators, identify accident trends, and present insights that can help users better understand road accident patterns in the United Kingdom during 2024.

3. Objectives

- To import and organize the UK Road Safety dataset in Microsoft Power BI.
- To clean and transform the data using Power Query Editor to ensure accuracy and consistency.
- To analyze the total number of road collisions, vehicles involved, and casualties.
- To examine the distribution of casualties based on severity levels such as fatal, serious, and slight injuries.
- To analyze road accidents under different weather conditions, light conditions, and road surface conditions.
- To identify the vehicle types most frequently involved in road collisions.
- To study the distribution of casualties across different age groups.
- To analyze monthly road collision trends and identify periods with higher accident occurrences.
- To design an interactive Power BI dashboard with filters and visualizations for easy exploration of the data.
- To generate meaningful insights that help users better understand road safety trends in the United Kingdom during 2024.

4. Dataset Description

The dataset used in this project contains information related to road safety and traffic accidents in the United Kingdom (UK) for the year 2024. The dataset was provided by the faculty for academic analysis and was available in Microsoft Excel (.xlsx) format.

The dataset includes details about road collisions, casualties, vehicles involved, weather conditions, road surface conditions, light conditions, casualty severity, and age groups. These attributes provide a comprehensive view of the factors associated with road accidents and help in identifying accident patterns and trends.

The dataset was imported into Microsoft Power BI Desktop, where it was cleaned and transformed using Power Query Editor before creating interactive visualizations and dashboards.

Dataset Information

S.No	Attribute	Description
1	Dataset Name	UK Road Safety Dataset
2	Country	United Kingdom (UK)
3	Year	2024
4	Source	Dataset provided by the faculty
5	File Format	Microsoft Excel (.xlsx)

S.No	Attribute	Description
6	Analysis Tool	Microsoft Power BI Desktop
7	Data Cleaning Tool	Power Query Editor

5. Methodology

The methodology adopted for this project involved a systematic process of collecting, preparing, analyzing, and visualizing the road safety data using Microsoft Power BI. The primary objective was to transform the raw dataset into an interactive dashboard that provides meaningful insights into road accidents in the United Kingdom (UK) for the year 2024.

The methodology followed in this project consists of the following stages:

Step 1: Dataset Collection

The UK Road Safety dataset for the year 2024 was provided by the faculty in Microsoft Excel (.xlsx) format. The dataset contained information related to road collisions, casualties, vehicle types, weather conditions, road surface conditions, light conditions, and age groups.

Step 2: Data Import

The Excel dataset was imported into Microsoft Power BI Desktop using the Get Data option. After importing, the dataset was examined to understand its structure, data fields, and overall quality before beginning the analysis.

Step 3: Data Cleaning and Transformation

The imported dataset was cleaned and transformed using Power Query Editor. The data cleaning process included:

- Checking for missing or null values.
- Verifying and correcting data types.
- Renaming columns for better readability.
- Removing unnecessary or duplicate records (if any).
- Ensuring the dataset was consistent and ready for analysis.

These steps improved the quality and reliability of the data used for visualization.

Step 4: Data Analysis

After cleaning, the dataset was analyzed to identify important trends and patterns. Key metrics such as total collisions, total vehicles involved, total casualties, fatal casualties, serious casualties, and slight casualties were calculated. Additional analysis was performed based on:

- Monthly collision trends
- Weather conditions
- Light conditions
- Road surface conditions
- Vehicle types
- Casualty age groups

Step 5: Dashboard Development

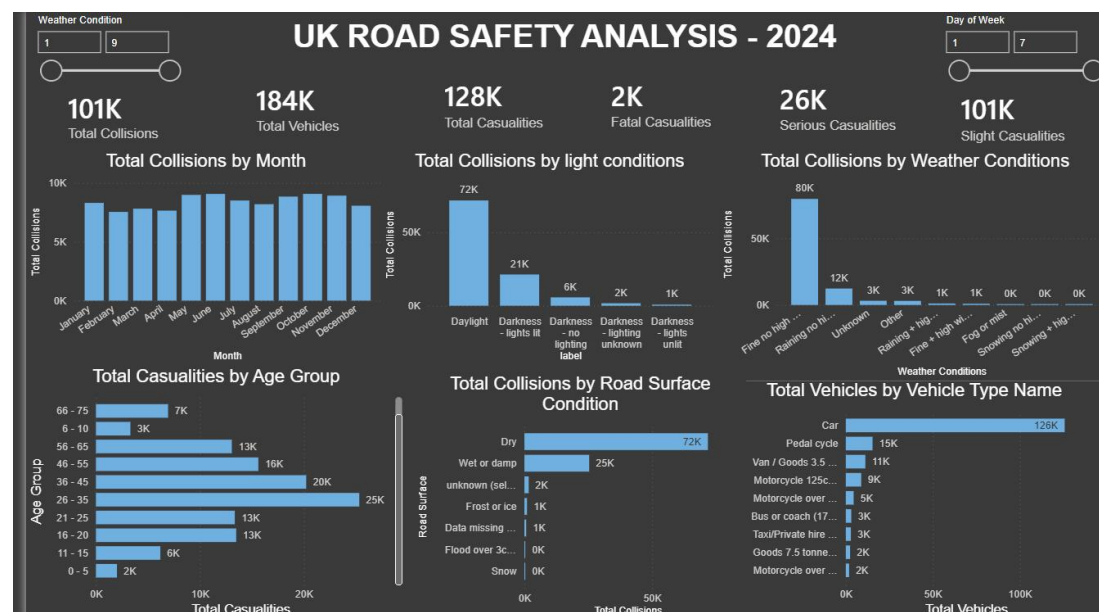
An interactive dashboard was designed in Microsoft Power BI using various visualizations such as KPI cards, bar charts, column charts, line charts, and slicers. These visuals were organized to present the information in a clear and user-friendly manner. Interactive filters were also added to enable users to explore the data from different perspectives.

Step 6: Insight Generation

The final dashboard was used to interpret the results and identify significant patterns in the road safety data. The insights obtained from the analysis helped in understanding accident trends, casualty distribution, and the influence of factors such as weather, lighting conditions, road surface conditions, and vehicle types on road accidents.

6. Data Analysis and Interpretation

The cleaned dataset was analyzed using Microsoft Power BI Desktop to identify road accident patterns and trends in the United Kingdom (UK) for the year 2024. Interactive visualizations were developed to examine key performance indicators, casualty severity, vehicle involvement, weather conditions, light conditions, road surface conditions, age groups, and monthly collision trends. The following interpretations were derived from the dashboard.



6.1 Overall Road Safety Statistics

The dashboard presents six Key Performance Indicators (KPIs) that provide an overview of road safety statistics.

KPI	Value
Total Collisions	101K
Total Vehicles	184K
Total Casualties	128K
Fatal Casualties	2K
Serious Casualties	26K
Slight Casualties	101K

Interpretation

The analysis shows that 101,000 road collisions occurred during the selected period, involving 184,000 vehicles and resulting in 128,000 casualties. Among these casualties, 101,000 were slight injuries, 26,000 were serious injuries, and 2,000 were fatal cases. This indicates that although road accidents occur frequently, the majority result in slight injuries, while fatal accidents represent a relatively small proportion of the total casualties.

6.2 Monthly Collision Trend

The line chart illustrates the variation in road collisions across different months.

Interpretation

The number of collisions gradually increases from the beginning of the year and reaches its highest levels during **October and November**. A slight decline is observed in **December**. This trend suggests that road accidents become more frequent during the latter half of the year, possibly due to increased traffic movement and seasonal factors.

6.3 Casualties by Light Conditions

The dashboard analyzes accidents under different lighting conditions.

Light Condition	Collisions
Daylight	72K
Darkness (Lights Lit)	21K
Darkness (No Lighting)	6K
Darkness (Lighting Unknown)	2K
Darkness (Lights Unlit)	1K

Interpretation

The highest number of collisions (72K) occurred during daylight. Comparatively fewer accidents occurred under different darkness conditions. This suggests that the higher volume of vehicles on the road during daytime contributes to a greater number of accidents than reduced visibility alone.

6.4 Casualties by Weather Conditions

The dashboard categorizes accidents according to weather conditions.

Interpretation

Most road collisions occurred during fine weather without high winds, accounting for approximately 80K collisions. Rainy conditions contributed to fewer accidents compared to fine weather. This indicates that traffic density and driver behavior may have a greater influence on accident frequency than weather conditions alone.

6.5 Casualties by Age Group

The dashboard displays the distribution of casualties across different age groups.

Interpretation

The 26–35 years age group recorded the highest number of casualties, followed by the 36–45 years age group. Younger and middle-aged adults are therefore more frequently involved in road accidents, likely due to higher daily travel for work, education, and personal activities.

6.6 Road Surface Condition Analysis

Road accidents were analyzed based on the condition of the road surface.

Interpretation

The majority of collisions occurred on dry roads, while wet or damp roads accounted for the second-highest number of accidents. This suggests that accident frequency is more closely related to traffic volume than road surface condition alone.

6.7 Vehicle Type Analysis

The dashboard presents the number of vehicles involved in road accidents by vehicle type.

Interpretation

Cars accounted for the highest number of vehicles involved in road collisions, followed by pedal cycles, vans, and motorcycles. Since cars are the most commonly used mode of transportation, they naturally represent the largest share of accident involvement.

6.8 Dashboard Filters

The dashboard includes interactive filters for:

- Road Surface Condition
- Weather Condition
- Light Condition

Interpretation

These filters enable users to analyze road accidents under specific conditions and compare accident patterns across different scenarios. The interactive design improves the usability of the dashboard and supports detailed exploration of the data.

7. Findings

Based on the analysis of the UK Road Safety Dataset (2024) using Microsoft Power BI, the following key findings were identified:

- A total of 101,000 road collisions were recorded, involving approximately 184,000 vehicles and resulting in 128,000 casualties.
- The majority of casualties were classified as slight injuries (101K), while 26K were serious injuries and 2K were fatal casualties. This indicates that although road accidents are frequent, fatal incidents represent a small proportion of the total casualties.
- The highest number of road collisions occurred during daylight (72K), suggesting that increased traffic volume during daytime contributes more to accidents than reduced visibility at night.
- Most accidents were recorded during fine weather without high winds (80K). This shows that favorable weather conditions do not necessarily reduce accident occurrences, as traffic volume is generally higher during such conditions.

- Analysis based on road surface conditions revealed that the majority of accidents occurred on dry roads (72K), followed by wet or damp roads (25K).
- The 26–35 years age group recorded the highest number of casualties, followed by the 36–45 years age group, indicating that working-age adults are more frequently involved in road accidents.
- Among all vehicle categories, cars were involved in the highest number of collisions, making them the most frequently affected vehicle type.
- The monthly trend analysis showed that road collisions gradually increased during the year and reached their highest levels in October and November, followed by a slight decline in December.
- The interactive dashboard developed in Microsoft Power BI enabled efficient analysis through filters for weather conditions, road surface conditions, and light conditions, making it easier to explore accident patterns from different perspectives.
- Overall, the dashboard successfully transformed raw road safety data into meaningful visual insights, helping users understand accident trends, casualty distribution, and the impact of different environmental and traffic-related factors on road safety in the United Kingdom during 2024.

8. Conclusion

The UK Road Safety Analysis Dashboard – 2024 was successfully developed using Microsoft Power BI Desktop to analyze and visualize road accident data in an interactive and meaningful manner. The project demonstrated how business intelligence tools can transform raw data into informative dashboards that support effective analysis and decision-making.

The dataset, provided in Microsoft Excel (.xlsx) format, was cleaned and transformed using Power Query Editor before being analyzed through various Power BI visualizations. The dashboard presented key performance indicators, including total collisions, total vehicles involved, total casualties, and casualty severity, along with detailed analyses based on monthly trends, weather conditions, light conditions, road surface conditions, vehicle types, and age groups.

The analysis revealed that the majority of road accidents resulted in slight injuries, while cars were the most frequently involved vehicle type. Most collisions occurred during daylight, under fine weather conditions, and on dry road surfaces. Additionally, individuals aged 26–35 years experienced the highest number of casualties, and collision trends increased during the latter half of the year, particularly in October and November.

Overall, the project highlights the effectiveness of Microsoft Power BI as a business intelligence tool for analyzing large datasets and presenting information through interactive dashboards. The insights obtained from this analysis provide a better understanding of road safety patterns in the United Kingdom during 2024 and demonstrate how data visualization can support data-driven decision-making. This project also enhanced practical skills in data cleaning, data transformation, dashboard design, and data interpretation using Power BI.